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| | **GLOBAL ECONOMICS**<br>**Grade: 11th - Learning Evidence 3**<br>**Statistical models and research planning — Version 2**<br>**Teacher: Nicolás López Cuéllar** | **1st TERM**<br>**2026–2027** |

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| **Learning objective:** Apply continuous probability models and sampling distributions to investigate economic and financial questions, justify statistical methods, and communicate evidence-based conclusions. | **Assessment criteria:**<br>C8: Uses Student's t-distribution tables, degrees of freedom, and one-tailed or two-tailed areas to determine critical values, central probabilities, and tail probabilities or probability ranges.<br>C9: Calculates and compares critical values and probability areas from the standard normal and Student's t-distributions, selecting and justifying the appropriate distribution according to whether the population standard deviation is known or estimated from the sample.<br>C10: Integrates continuous probability modelling, simulation, sampling analysis, and portfolio-risk measures to investigate an economic or financial question, justifying the selected methods and communicating results, assumptions, limitations, and evidence-based conclusions. |

**Name:** _____ **Grade:** _____ **Date:** _____

## Criteria assessment

Partial credit may be awarded when the correct procedure is clearly demonstrated, even if arithmetic errors are made. No points will be awarded for answers without a demonstrated procedure. A minimum of **4 points** is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

| № | C8 |
|---|---|
| 1–5 | 5 |
| 6–10 | 5 |
| **Total** | **10** |

| № | C9 |
|---|---|
| 11–13 | 6 |
| 14–16 | 6 |
| **Total** | **12** |

## Conventions

**Models.** C8 (1–10): $X = m + aT$. C9 (11–16): $X_T = m + aT$, $X_N = m + aZ$, where $T \sim t_{n-1}$, $Z \sim N(0,1)$, $n$ is sample size, $m$ is location, and $a > 0$ is scale.

**Standardization.** Subtract $m$ and divide by $a$:

$$T = \frac{X_T - m}{a}, \qquad Z = \frac{X_N - m}{a}.$$

For inverse problems, transform back using $x = m + at$ or $x = m + az$.

**C9 comparisons.** For every problem, calculate $r_T$, $r_N$, and

$$E_A = |r_T - r_N|, \qquad E_{Pt} = \frac{100E_A}{|r_T|}, \qquad E_{Pn} = \frac{100E_A}{|r_N|}.$$

## 1. Warehouse energy deviations [C8-1]

**C8 – Direct interval above the mean.** Over $n = 18$ days, electricity-use deviation from forecast, in kWh, has mean $m = 2.4$ and approximate standard deviation $a = 7$. Use $X = 2.4 + 7T$, $T \sim t_{n-1}$. Find $P(8.441 < X < 14.58)$. **[1]**

## 2. Retail revenue alerts [C8-1]

**C8 – Direct unequal combined tails.** Over $n = 8$ weeks, revenue deviation from plan, in thousands of dollars, has mean $m = 1.6$ and approximate standard deviation $a = 5$. Use $X = 1.6 + 5T$, $T \sim t_{n-1}$. Find the probability of an alert: $X < -7.875$ or $X > 8.675$. **[1]**

## 3. Courier arrival deviations [C8-1]

**C8 – Direct left tail.** A sample of $n = 11$ courier trips has arrival-time deviation from schedule, in minutes, with mean $m = 1.2$ and approximate standard deviation $a = 3.5$. Use $X = 1.2 + 3.5T$, $T \sim t_{n-1}$. Find $P(X < -5.142)$. **[1]**

## 4. Freight cost deviations [C8-1]

**C8 – Direct right tail.** A sample of $n = 14$ consignments has cost deviation from budget, in dollars, with mean $m = -5$ and approximate standard deviation $a = 9$. Use $X = -5 + 9T$, $T \sim t_{n-1}$. Find $P(X > -12.83)$. **[1]**

## 5. Crop shipment deviations [C8-1]

**C8 – Direct interval below the mean.** A sample of $n = 9$ crop shipments has mass deviation from forecast, in tonnes, with mean $m = -1.2$ and approximate standard deviation $a = 3.6$. Use $X = -1.2 + 3.6T$, $T \sim t_{n-1}$. Find $P(-9.5016 < X < -4.4004)$. **[1]**

## 6. Package mass alerts [C8-1]

**C8 – Inverse equal-tail alert limits.** A sample of $n = 10$ packages has mass deviation from target, in grams, with mean $m = -0.5$ and approximate standard deviation $a = 1.4$. Use $X = -0.5 + 1.4T$, $T \sim t_{n-1}$. Find two alert limits leaving a combined outside probability of 0.05, split equally. **[1]**

## 7. Operating cost cutoff [C8-1]

**C8 – Inverse lower-tail cutoff.** Over $n = 16$ produc-

tion days, operating-cost deviation from budget, in dollars, has mean $m = -3.5$ and approximate standard deviation $a = 6$. Use $X = -3.5 + 6T$, $T \sim t_{n-1}$. Find $x$ such that $P(X < x) = 0.10$. **[1]**

### 8. Dispatch time alerts [C8-1]

**C8 – Inverse unequal-tail alert limits.** A sample of $n = 17$ dispatches has departure-time deviation from schedule, in minutes, with mean $m = -2.5$ and approximate standard deviation $a = 7$. Use $X = -2.5 + 7T$, $T \sim t_{n-1}$. Find $L, U$ with $P(X < L) = 0.01$ and $P(X > U) = 0.10$. **[1]**

### 9. Loading time cutoff [C8-1]

**C8 – Inverse upper-tail cutoff.** A sample of $n = 22$ loading jobs has duration deviation from target, in minutes, with mean $m = 2.5$ and approximate standard deviation $a = 8$. Use $X = 2.5 + 8T$, $T \sim t_{n-1}$. Find $x$ such that $P(X > x) = 0.025$. **[1]**

### 10. Demand deviation interval [C8-1]

**C8 – Inverse central interval.** Over $n = 13$ weeks, demand deviation from forecast, in tonnes, has mean $m = 5$ and approximate standard deviation $a = 10$. Use $X = 5 + 10T$, $T \sim t_{n-1}$. Find $L < U$ enclosing the central 90% of deviations with equal excluded tails. **[1]**

### 11. Delivery schedule deviations [C9-2]

**C9 – Direct right tail.** A sample of $n = 13$ deliveries has arrival-time deviation from schedule, in minutes, with mean $m = 3.2$ and approximate standard deviation $a = 4.5$. Compare $X_T = 3.2 + 4.5T$, $T \sim t_{n-1}$, with $X_N = 3.2 + 4.5Z$. Find $P(X > -0.7285)$. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$. **[2]**

### 12. Fuel forecast deviations [C9-2]

**C9 – Direct interval crossing the mean.** Over $n = 14$ operating days, fuel-use deviation from forecast, in litres, has mean $m = 1.2$ and approximate standard deviation $a = 7$. Compare $X_T = 1.2 + 7T$, $T \sim t_{n-1}$, with $X_N = 1.2 + 7Z$. Find $P(-8.25 < X < 16.32)$. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$. **[2]**

### 13. Sales revenue shortfalls [C9-2]

**C9 – Direct left tail.** Over $n = 11$ weeks, sales-revenue deviation from target, in thousands of dollars, has mean $m = -2.5$ and approximate standard deviation $a = 6$. Compare $X_T = -2.5 + 6T$, $T \sim t_{n-1}$, with $X_N = -2.5 + 6Z$. Find $P(X < -13.372)$. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$. **[2]**

### 14. Cargo mass deviation interval [C9-2]

**C9 – Inverse central interval.** A sample of $n = 11$ cargo loads has mass deviation from target, in kg, with mean $m = 1.8$ and approximate standard deviation $a = 5$. Compare $X_T = 1.8 + 5T$, $T \sim t_{n-1}$, with $X_N = 1.8 + 5Z$. Find the central 95% interval with equal excluded tails under each model. Calculate $E_A$, $E_{Pt}$, and $E_{Pn}$ separately for each endpoint. **[2]**

### 15. Production cost savings [C9-2]

**C9 – Inverse lower-tail cutoff.** A sample of $n = 9$ batches has cost deviation from budget, in dollars, with mean $m = -1.3$ and approximate standard deviation $a = 3$. Compare $X_T = -1.3 + 3T$, $T \sim t_{n-1}$, with $X_N = -1.3 + 3Z$. Find the cutoff $x$ below which 5% of deviations fall. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$. **[2]**

### 16. Order processing cutoff [C9-2]

**C9 – Inverse upper-tail cutoff.** A sample of $n = 18$ orders has processing-time deviation from target, in minutes, with mean $m = 2.4$ and approximate standard deviation $a = 4$. Compare $X_T = 2.4 + 4T$, $T \sim t_{n-1}$, with $X_N = 2.4 + 4Z$. Find the cutoff $x$ exceeded by 10% of deviations. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$. **[2]**

### 17. Research option [C10-6]

**Exam:** Select one model and describe plausible real-life data in 1–2 sentences. **[6]**

- **(a)** Linear probability density over a finite interval.
- **(b)** Exponential distribution (waiting times).
- **(c)** Normal distribution (measurements).

**Online report:** Submit a PDF through **Open** using real economic or financial data and the selected model. Include simulation, sampling, portfolio risk, methods, assumptions, limitations, and evidence-based conclusions. The report is the C10 evidence.